Face Mask Classification Using MobileNetV2

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Course: Computer Vision Capstone
GitHub: https://github.com/Zachrist321/facemask_classifier_mbnv2
Live App: https://facemask-classifier-mbnv2.streamlit.app
Dataset: MAKS Face Mask Dataset (Kaggle)

1. Introduction

Public health monitoring during respiratory disease outbreaks requires automated tools to detect whether people wear face masks correctly. Manual inspection does not scale in crowded environments such as hospitals, airports, and retail spaces.

This project builds a deep learning image classification system that identifies mask compliance from face images. We use transfer learning with MobileNetV2 pretrained on ImageNet, fine-tuned on the MAKS dataset. The final deliverable is a Streamlit web application that accepts uploaded photos or webcam input and returns a class prediction with confidence scores.

2. Problem Statement

Given a face image, the system will classify mask compliance into three categories so that public-health and security teams can monitor mask usage in real time.

Classes:

1. with_mask - face covered correctly
2. mask_weared_incorrect - mask present but worn incorrectly
3. without_mask - no mask detected

3. Methodology

3.1 Dataset

Source: MAKS (Kaggle) - 853 images with Pascal VOC XML bounding-box annotations.
Preprocessing: Each annotated face region was cropped with 15% margin and resized to 224x224 pixels, yielding 4,072 face samples.
Split: 80/20 stratified per class (seed=42). Train: 3,259 | Validation: 813.

3.2 Model

Architecture: MobileNetV2 (ImageNet weights) with frozen backbone.
Trainable head: GlobalAveragePooling -> Dropout(0.3) -> Dense(3, softmax).
Only the classification head was trained (~0.3% of parameters) for fast GPU training.

3.3 Training

Platform: Google Colab (T4 GPU).
Optimizer: Adam, lr=1e-3 | Batch size: 32 | Epochs: 15 (early stopping).
Augmentation: random flip, brightness, contrast.
Best validation accuracy: 92.5%.

3.4 Evaluation Metrics

Accuracy, per-class Precision/Recall/F1-score, and confusion matrix on validation set.

4. Proposed Solution

A transfer-learning pipeline: (1) parse VOC annotations and crop faces, (2) train a MobileNetV2 classifier with frozen backbone, (3) evaluate on held-out validation crops, (4) deploy via Streamlit for upload/webcam inference.

5. System Architecture

Components:
- Data layer: MAKS images + XML annotations -> face crops
- Model layer: MobileNetV2 + custom softmax head
- Inference layer: TensorFlow prediction on 224x224 RGB input
- UI layer: Streamlit app (upload or webcam)
- Hosting: Streamlit Community Cloud linked to GitHub repository

6. Implementation Details

Languages/Libraries: Python 3.11, TensorFlow 2.15, Streamlit, Pillow, scikit-learn.
Hardware (training): Google Colab T4 GPU.
Hardware (deployment): Streamlit Cloud CPU.
Repository structure: streamlit_app.py (UI), src/ (data & model utilities), face_mask_artifacts/ (trained weights), notebooks/ (Colab training notebook).

7. Results

7.1 Overall Performance

Validation accuracy: 92.5%
Weighted F1-score: 0.91

7.2 Per-Class Results

with_mask: Precision 0.94 | Recall 0.97 | F1 0.96 (646 samples)
without_mask: Precision 0.86 | Recall 0.85 | F1 0.85 (143 samples)
mask_weared_incorrect: Precision 0.50 | Recall 0.08 | F1 0.14 (24 samples)

7.3 Failure Cases

Four of five documented failures involved mask_weared_incorrect predicted as with_mask. Causes: (1) class imbalance - only 99 training samples for incorrect mask vs 2,586 for with_mask; (2) visually ambiguous partial-mask faces; (3) one without_mask case where shadow/hand near face resembled a mask edge.

7.4 Limitations

Model trained on cropped faces - full-scene photos without clear faces perform poorly. Incorrect-mask class needs more training data. iPhone HEIC files require conversion before upload.

8. Deployment

The application is deployed on Streamlit Community Cloud:
https://facemask-classifier-mbnv2.streamlit.app

Source code:
https://github.com/Zachrist321/facemask_classifier_mbnv2

Users upload a face photo or use a webcam. The app displays the predicted class and probability bars for all three classes.

9. Conclusion

We built a working face mask classifier achieving 92.5% validation accuracy using MobileNetV2 transfer learning. The system correctly identifies with_mask and without_mask in most cases. The mask_weared_incorrect class remains challenging due to limited data. Future work: collect more incorrect-mask samples, add face detection for full-scene images, and fine-tune the last MobileNet blocks.

10. References

[1] MAKS Dataset - Kaggle: Face Mask Detection Dataset
[2] Howard et al., MobileNetV2, arXiv:1801.04381
[3] TensorFlow Documentation - Transfer Learning
[4] Streamlit Documentation - Community Cloud Deployment
[5] Project II Guidelines - DNN Based Computer Vision Capstone